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PROJECT PHASE 2

Ecosystem Assessment of Keta and Songor Ramsar sites

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Acronyms

DO	Dissolved Oxygen
GIS	Geographical Information System
K	Potassium
KLRS	Keta Lagoon Complex Ramsar Site
ORP	Oxidation Reduction Potential
N	Nitrate
P	Potassium
TDS	Total Dissolved Solids
UNESCO	United Nations Education, Scientific and Cultural Organisation
WEH	Wetland Ecosystem Health

Executive Summary

Wetlands are among the most productive ecosystems globally, providing vital services such as biodiversity support, water filtration, and carbon storage. Along Ghana's coast, lagoons like the Keta Lagoon Complex and the Songor Ramsar Site hold immense conservation value and have received both local and international recognition. However, these ecosystems face increasing threats from population growth, resource exploitation, and climate variability.

This assessment was conducted in December 2025 to evaluate ecosystem health at the two Ramsar sites and to generate scientific data that can guide sustainable management and long-term monitoring. Using supervised GIS mapping and a stratified random sampling approach, 65 sampling sites were established across diverse ecosystem types. In total, 270 water quality and 140 soil samples were collected at Keta, while 117 water quality and 28 soil samples were gathered at Ada. The exercise also built local capacity by training community change agents in data collection.

Analysis of physico-chemical and soil indicators revealed significant spatial variability in water quality and nutrient levels. Soil parameters such as moisture, humus, and nutrient concentrations varied across sites, reflecting ecological differences. Notably, a general temperature increase of 3.2–6.1 °C was observed, with higher oxidation-reduction potential (ORP) values recorded in Ada. pH levels at Keta ranged between 6.5 and 7.5, while potassium concentrations were consistently elevated across sampling areas. Despite seasonal fluctuations, both wetlands remain productive, though their resilience is increasingly challenged by hydrological changes and habitat degradation.

The findings underscore the urgent need for continued monitoring and adaptive management to safeguard the ecological integrity and ecosystem services of Ghana's coastal wetlands.

Introduction

Assessing ecosystem health is a priority for governments, scientists and managers working in the field of environmental conservation. The scientific literature is replete with discussions on ecosystem health and approaches to assess it, with applications to aquatic and terrestrial environments incorporating economic, environmental and social processes (Allyson et al., 2016). Ecosystem-scale management is increasingly being initiated around the world to cope with complex problems spanning diverse environmental attributes over large areas. Assessing impacts of ecosystem scale change will commonly require a broad scope in space, ecosystem features of public interest and different kinds of information (Bain et al., 2008).

Environmental systems are considered benefitting from wetlands through protein production, water sanctification, groundwater recharge, climate purification, nutrient cycling, flood mitigation and biodiversity preservation (Keddy, 2000). Wetlands are among the most productive ecosystems on the planet, playing crucial roles in biodiversity, water filtration and carbon sequestration. They are vital ecosystems that provide numerous ecological, hydrological, and socio-economic benefits. They are characterized by the presence of water, which plays a fundamental role in shaping their structure and function (Comegna, 2024). Healthy wetland ecosystems are essential for a resilient society and a sustainable economic life. Productive soils promote agricultural activities and income of peasant farmers. Well maintained rivers and lakes supply portable water, suitable habitats for fish and wildlife and provide opportunities for recreation. Besides direct economic benefits healthy wetland ecosystems are essential providers of many ecosystem regulating services. Achieving lasting social, environment and economic development thus requires the sustained provision of these services, which are derived from functional and healthy ecosystems (Comegna, 2024).

Notwithstanding their benefits, wetland ecosystems have witnessed a rapid detraction, including a reduction in area, deterioration in water quality and loss of biodiversity. The combined processes of wetland degradation are seen in the structural and functional disorder of the wetlands. Wetland ecosystems have faced several difficulties due to intensive economic development, rapid population growth, climatic variabilities, ecological fragmentation, over-exploitation of already stressed natural resources, and encroachment driven by poverty in most communities.

Globally 64% of the wetlands have disappeared since 1970 (Venkatesan et. al 2022). With 40% of the world's species reliant in some way on wetlands, the loss of these valuable ecosystems is forcing many species to the brink of extinction. Eighty three percent (83%) of freshwater species are in decline worldwide whiles 80% of our global wastewater is released into wetlands untreated (WWT, 1921). Pollution from factories, fertilizers, pesticides or from major spills, all pose serious threats to wetlands (WWT, 1921).

Assessments are the basis for evaluating problems and identifying the best-suited solutions. Conducting assessments thus play several roles and functions in relation to policy, science, and decision making (UNEP, 2015). To achieve wise use and for wetlands to contribute fully to sustainable development, policy makers and practitioners need to recognize the important functions and the multiple values of wetlands and reflect them in their decisions, policies, and actions (Kumar et. al, 2017).

Without wetlands, the water cycle, carbon cycle and nutrient cycle would be significantly altered, mostly detrimentally. Yet, often due to a failure to recognize these multiple,

interconnected values, policies and decisions do not sufficiently consider these interconnections and interdependencies (Russi et. al, 2013). The Ramsar Convention has recognized the need to integrate the important functions and multiple values of wetlands into decision making and has produced policy briefs, technical reports, and wider guidance to address the importance of this issue (De Groot et. al, 2006).

Rapport et al. (1985), popularized the word "ecosystem health". They characterized it as a form of biological system dependability and maintainability, which has the capacity for keeping up its organizational structure as well as natural regulating and recuperation capacity after resilience. The Millennium Ecosystem Assessment has defined ecosystem condition as the capacity of an ecosystem to deliver ecosystem services, relative to its potential capacity (Millennium Ecosystem Assessment, 2005).

The concept of ecosystem condition is linked to well-being through ecosystem services. Ecosystems need to be in good condition to provide a set of essential services which, in turn, deliver benefits and increase well-being. Drivers of change can have a positive (e.g. conservation) or negative (pressures) impact on ecosystem condition.

The coastline of Ghana stretches over 550 km between Togo and Cote d'Ivoire. It has important features such as lagoons that provide diverse ecological values and benefits to adjacent communities and these include fishing, salt extraction, water transport and reeds extraction. Some of the lagoons along the Ghanaian coast have received both local and international recognition due to the immense conservation values. However, it is evident that the values and benefits provided by the lagoons are under increasing threat that ranges from resource overexploitation and degradation due to population expansion.

This report provides a description of the current ecosystem status of the Keta and Songor lagoons which are internationally important wetlands and have been designated as Ramsar sites. The evaluation of the wetland environment was done by using techniques that combine quantitative and qualitative assessment methods.

The main objective of this study was to establish the status of the wetland ecosystem health (WEH) on the basis of available data, to categorize the health conditions and identify the habitats where the ecosystem health condition is deteriorating or improving.

2.0 Materials and methods

2.1 Study area

The two sites presented in this report are among the six designated Ramsar sites in Ghana, five of which are coastal wetlands and one is inland. The five coastal wetlands cut across three (3) coastal regions, namely the Volta, Greater Accra and Central Regions.

2.1.1 Keta Lagoon Complex Ramsar site

The Keta Lagoon Complex Ramsar Site (KLRS) is situated in the Southern part of the Volta Region. The Wetland is located between latitudes 5° 45' N and 6° 05' N and Longitudes 0° 50'E and 1° 08' E. It is bordered on the west by the Volta River, in the south and east by the Gulf of Guinea and in the north by the Accra-Lome highway.

The Ramsar Site comprises a large expanse of open lagoon with brackish water, flood plain, marshland and extensive mangrove stands. The Keta lagoon complex, which is generally close to the sea, is interconnected to the Avu lagoon located in the northern part and also to the delta of the Volta River through series of inter-connecting streams. The open water of Keta Lagoon is estimated to cover an area of 300 km^2 which varies with the season. The site covers 530 Km^2 and stretches for 40 Km along the coast. It is separated from the sea by a narrow coastal ridge which is only 2.5 Km at its widest point and 0.92 km at the narrowest portion. The total catchment area feeding the lagoon is about 2,900 m^2 and was designated as a Ramsar site in 1992. The construction of the Akosombo dam has been blamed for a number of ecologic changes which have occurred within the site in the recent past. These include silting and blockage of the channels interconnecting the lagoons which is thought to disrupt the direct and natural drainage system that used to regulate the lagoon water levels. The lagoon exhibits significant spatial and temporal variability in environmental conditions. Salinity is the most variable parameter reflecting different river run-off and evaporation conditions at different parts of the lagoon (Tuffuor, 1999).

2.1.2 Songor Ramsar site and UNESCO Biosphere Reserve.

Located in the Greater Accra Region, Songor Ramsar site is the second largest wetland in Ghana. It lies between latitudes 06.50N and 00°30E (Wetland Management (Ramsar Sites) Regulations, 1999), and covers an area of 51,133 hectares (Ofori-Danson, 1999). It supports the existence of rich and diverse ecosystems such as riverine, estuarine, floodplains, fringing forests, and inshore coastal waters. The wetland constitutes the western portion of the Volta Estuary thus, providing good location for tourism development. The site falls in a low-lying area (the highest point being less than 10m above sea level); this serves as a flood plain and a storm buffer and supports the communities from getting flooded (Ofori-Danson, 1999). The site comprises of three main habitat types:

(i) a brackish water ecosystem comprising the Songor lagoon which is one of the two lagoon systems associated with the Volta River Estuary (the other is Keta lagoon) and its marshes and mudflats. The lagoon is closed off from the sea (Mensah, 1979) and is situated to the west of the Volta Estuary. The land around the lagoon is low-lying, with the highest point less than 10m above sea level. The general elevation does not exceed 75 m above sea level in the northern part and 15 m above sea level near the coast. The open water covers an estimated area of 115 km^2 and extends about 20 km^2 along the coast and approximately 8 km inland behind a narrow sand dune on which villages like Pute and Totokpoe are situated. There are other small lagoons in the site that include Kadza, Kunye and Truku.

(ii) a riverine ecosystem comprising the lower Volta River, interconnected creeks, flood plains with degraded mangroves.

(iii) a marine ecosystem with a broad sandy beach in the south and coastal savannah vegetation.

2.2 Methodology

The study involved a systematic approach to data collection to obtain a set of wetland health indicators. The data captured include variable chemical and physical water quality and soil characteristics of the different wetland ecosystem types. Indicators of degradation and threats to the ecosystem were Identified. A supervised GIS map prepared for the study areas provided the necessary classification for sampling sites for data collection.

The sampling sites were determined in consultation with the community change agents, taking into consideration the diverse ecosystem, notably the aquatic and the terrestrial components. Sampling of each lagoon and the surrounding wetland vegetation was based on selected stations in the prepared maps. Since Keta was larger, sampling included large areas of the surrounding wetlands and also sections of the open water.

The stations mostly corresponded to the labelled sites on the GIS map. The aquatic samples were randomly collected using dugout canoes. Aquatic areas limited by vegetation were waded through and samples were collected close to the terrestrial area. A set of nine (9) different water quality and seven (7) soil parameters were collected in 65 sampling stations within Keta and Ada sampling areas.

Within the Keta Lagoon Complex site, 270 water quality and 140 soil samples were collected in the different ecosystem types whiles 117 water quality and 28 soil parameters were gathered in Ada. The data was collected in December 2025 using stratified random sampling method. Microsoft Office (Excel spreadsheet) was used to compute and analyse physico-chemical parameters obtained from the field survey. The results obtained was interpreted using descriptive statistics. Graphs and appropriate tables were developed using Microsoft Office (Excel). Analysis of Variance (ANOVA) was used for comparison of means of multiple variables.

2.2.1 Physico-chemical parameters

Data was gathered from a total of 387 random sampling stations, comprising 117 and 270 sampling stations in Ada and Keta Ramsar sites, respectively. At each sampling stations the following parameters were measured: pH, conductivity, ORP (oxidation-reduction potential), total dissolved solids (TDS), salinity, resistivity, dissolved oxygen in concentration and percentage, temperature using Apera SX751 water testing device, which provided the result of the field data *in situ*. The data was collected in December, which was supposed to be the dry period in Ghana. However, adequate rains occurred during the sampling period and this was likely to influence the result of some physico-chemical parameters.

2.2.2 Soil samples

Soil samples were collected from a total of 168 sampling stations. Within the terrestrial sampling areas, 28 stations were designated in Ada and 140 in Keta Ramsar sites. Results of field data were obtained *in situ* using Portable Terminal 4002-BXSZD Ver 1.0, a professional handheld quick test platform.

2.2.3 Georeferenced positions of sampling stations

All sampling stations were georeferenced using a Garmin Inreach Explorer. The details are presented in Tables 1 to 4 below:

Table 1: Water sampling - sampling sites and description of sites, Keta.

Coordinates of sampling sites	Sampling sites	Description of sites
N 05° 47.012, E 000° 43.276	Anyanui	Anyanui landing site
N 05° 46.564, E 000° 43.921	Fuveme	Fuveme creek
N 05° 46.704, E 000° 42.476	Agorkedzi	Agorkedzi creek
N 05° 47.571, E 000° 43.140	Tunu	Tunu creek
N 05° 47.308, E 000° 43.600	Anyanui	Anyanui - Tunu road
N 05° 47.366, E 000° 43.609	Anyanui	Anyanui - Tunu road
N 05° 47.195, E 000° 45.235	Agbledome	Agbledome
N 05° 46.442, E 000° 46.910	Dzita	Dzita
N 05° 46.359, E 000° 47.451	Akplotorkor	Akplotorkor sites 1
N 05° 46.383, E 000° 48.621	Akplotorkor	Akplotorkor site 2
N 05° 46.957, E 000° 49.803	Saviatula	Saviatula Bridge
N 05° 48.761, E 000° 47.953	Salo	Salo road
N 05° 49.464, E 000° 47. 405	Salo	salo mangrove market
N 05° 51.700, E 000° 45.274	Agortoe	Agortoe site 1
N 05° 51.786, E 000° 45.176	Agortoe	Agortoe site 2
N 05° 51.977, E 000° 44.884	Agbatsivi	Agbastivi site 1
N 05° 52.124, E 000° 44.733	Agbastivi	Agbastivi site 2
N 05° 52.021, E 000° 42.560	Galotse	Galoste site 1
N 05° 52.818, E 000° 41.397	Galotse	Galoste site 2
N 05° 59.136, E 000° 50.586	Genui	Gehnui Road
N 06° 01.888, E 000° 58.854	Afiadenyigba	Afiadenyigba bridge
N 06° 01.997, E 000° 59.471	Afiadenyigba	Afiadenyigba road / lagoon
N 05° 58.118, E 000° 59.796	Havedzi	Havedzi bridge
N 05° 55.868, E 000° 59.397	Vodza	Lagoon view hotel
N 05° 50.019, E 000° 57.225	Woe	Woe landing bridge
N 05° 47.833, E 000° 43.869	Anloga	Anloga main market

Table 2: Water sampling - sampling sites and water parameters, Songor, Ada.

Coordinates of sampling area	Sampling site	Description of area
N 05° 46.259, E 000° 39.185	Azizanya	Azizanya-Kewunor bridge
N 05° 46.371, E 000° 38.447		Ayigbo bridge
N 05° 47.111, E 000° 34.209	Patukope	Patukope creek
N 05° 47.096, E 000° 33.014	Pute	Pute walkway lagoon intake
N 05° 47.013, E 000° 32.434		Lagoon underground seepage
N 05° 47.212, E 000° 37.253	Futuenya	Furuenya creek, bridge
N 05° 48.425, E 000° 36.491	Big Ada	Big Ada creek, bridge
N 05° 50.321, E 000° 33.316	Wasakuse	Wasakusewayo, Restoration site
N 05° 50.322, E 000° 33.237	Togbloku	Togbloku lagoon 1
N 05° 50.322, E 000° 33.237		Togbloku lagoon 2
N 05° 48.291, E 000° 25.396	Obane	Obane creek

Table 3: Soil sampling - sampling sites and description of sites, Keta.

Coordinates of sampling site	Sampling site	Description of site
N 05° 47.308, E 000° 43.600	Anyanui	Dead mangroves, Tunu road
N 05° 47.366, E 000° 43.609	Anyanui	Tunu road
N 05° 47.174, E 000° 45.226	Agbledorne	Agbledorne site 1
N 05° 47.146, E 000° 45.215	Agbledorne	Agbledorne site 2
N 05° 47.086, E 000° 45.176	Agbledorne	Agbledorne site 3
N 05° 46. 442, E 000° 46.910	Dzita	Dzita
N 05° 47. 014, E 000° 47.749	Saviatula	Saviatula bridge
N 05° 48. 761, E 000° 47.953	Salo	Salo road
N 05° 49. 466, E 000° 47.412	Salo	Salo mangrove market
N 05° 51. 700, E 000° 45.274	Agortoe	Agortoe
N 05° 52. 636, E 000° 43.788	Agbatsivi	Agbastivie
N 05° 52. 021, E 000° 42.560	Galotse	Galotse site 1
N 05° 51. 947, E 000° 42.269	Galotse	Galotse site 2
N 05° 52. 818, E 000° 41.392	Galotse	Galotse site 3
N 05° 49. 136, E 000° 50.586	Genui	Genui Road
N 06° 01. 600, E 000° 55.831	Aborlorvie	Upland fishing area
N 06° 01. 739, E 000° 55.767	Aborlorvie	further upland
N 06° 01. 001, E 000° 55.471	Aborlorvie	Proposed restoration site
N 06° 02. 297, E 000° 55.440	Aborlorvie	Proposed restoration site
N 06° 01. 888, E 000° 58.854	Afidenyigba	Afidenyigba bridge

Table 4: Soil sampling - sampling site and soil parameters, Songor, Ada.

Coordinates of sampling area	Sampling site	Description of area
N 05° 47.082, E 000° 32.585	Pute	Pute walkway, lagoon intake, black mangroves
N 05° 47.070, E 000° 33.027		Pute lagoon, white mangroves
N 05° 50.338, E 000° 33.322	Wasakuse	Wasakuse ngwa water near ponds
N 05° 48.338, E 000° 33.322	Obane	obane restored mangrove

3.0 Results and Discussion

3.1 Physical water parameters

3.1.1 pH: Keta

Acidity or alkalinity, expressed as pH, impacts an ecosystem, thus cellular respiration and homeostasis depend on optimal pH ranges. Generally, the mean pH value of water sampled across the sampling sites ranged between 6.8 and 7.96. However, clear variations were obvious among the sampling stations. This may have been influenced by the concentration of acid or base, temperature, presence of buffers and ionic strength.

At the Keta Lagoon Complex Rasmar site, the pH value at Anyanui–Dzita area ranged between 7.1 and 7.59. However, the value at Anyanui-Tunu road was 2.7, which was an exceptional occurrence that needs further studies and long-term monitoring. The condition impacted negatively on flora species especially Rhizhphora. A range of 6.5 to 7.6 was recorded within the sampling areas of Salo, Galosota and Genui area where there are well developed Rhizophora species. At Aborlorvi–Norpoli, the value ranged between 7.2 and 7.8. The values of the data sampled along the bank of the main lagoon from Afiadenyigba to Anloga was high (between 7.8 and 8.2), probably due to temperature variations and other physical and environmental factors.

3.1.2 pH: Songor, Ada

The mean pH value recorded at Songor Ramsar and Biosphere Reserve ranged between 7.3 and 7.76. The value recorded at Azizanya and Ayigbo close to the estuary ranged between 7.6 and 8.0, whiles Patukope-Pute was between 7.1 and 8.1, and in the Futue and the Luhue creeks it was between 8.1 and 8.6. The values recorded at Wasakuse, Toku lagoon and Obane creek varied between 7.3 and 8.1. The condition of these creeks within the site is critical for the sustenance and continual development of mangroves and other aquatic life.

The lack of patterns displayed in the values recorded during the period may probably be due to wind-induced mixing of the water to produce a homogeneous pH system. Also, the presence of some high buffers such as carbonates might have effectively resisted drastic pH changes.

3.2 Temperature

3.2.1 Keta

The mean temperature values recorded in the lagoons varied between 30.24°C and 31.3°C. This was probably influenced by the depth of the lagoon, the season, and the period the data was gathered. The temperature value in the Keta area was between 25°C and 30.3°C between dawn and just before noon. In the afternoon, a value between 31.19°C and 37.3°C was recorded. There was a temperature rise of 5.3°C just before noon and 6.11°C during noon. The temperature rise levels were probably influence by the sampling period.

3.2.2 Songor, Ada

The temperature values recorded for Songor, Ada, varied within the sampling sites with a mean value ranging from 31.2°C to-34.6°C. The temperature rise between the sampling period was gradual and lower. The value recorded in the morning was between 30.3 and 33.5°C, with 3.2°C temperature rise between dawn and noon. At noon, a value between 33.9°C and 37.2°C was recorded, with 3.3°C temperature rise during noon. Generally, the water temperature values recorded were optimal, a critical factor in determining the overall water quality. This significantly also influence the physical, chemical, and biological characteristics of the aquatic ecosystems.

3.3 Water chemical parameters: Conductivity, salinity, total dissolved solids, Resistivity, Dissolved oxygen, and Oxidation reduction potential.

3.3.1 Keta

Within the Keta Ramsar site, water chemical parameters recorded included fresh water and hyper-salinity conditions, typical of coastal lagoons in Ghana. The values were however influenced by the intermittent rains during the sampling period in the dry season in December. The highest value of conductivity (63.1 µs), total dissolved solids (88.1ppm) and salinity (40 ppt) occurred at Dzita. However, the values recorded for resistivity (Res), dissolved oxygen (DO) and oxidation reduction potential (ORP) were relatively moderate (15.8KΩ, 13.23 mg/l and 160 mV, respectively) for the same sampling site. Low ORP is a common characteristic of wetlands with waterlogged or anaerobic features influencing the availability of elements such as iron and manganese. The high salinity was probably from high evaporation and the presence of other ions in the water. The low salinity value of 4.3 ppt, conductivity value of 69.3 µs, TDS value of 100 ppm and ORP of 29 mV was probably influenced by the rainfall during the sampling period. High oxygen values at Anyanui, Agblodome, Apklotorkor, Saviatula and Salo (Figure 1) were probably influenced by windy conditions, shallow water depth, high water temperatures and swift flow of the water. Salinity values were higher in the lagoons, some creeks and occasionally closed water systems. Sites such as Anyanui, Tunu, Akplortorkor, Saviatula, and Genui had salinity values that ranged between 10.6 and 40 ppt (Figure 2). The sites were observed to support the development of mangroves and other aquatic life.

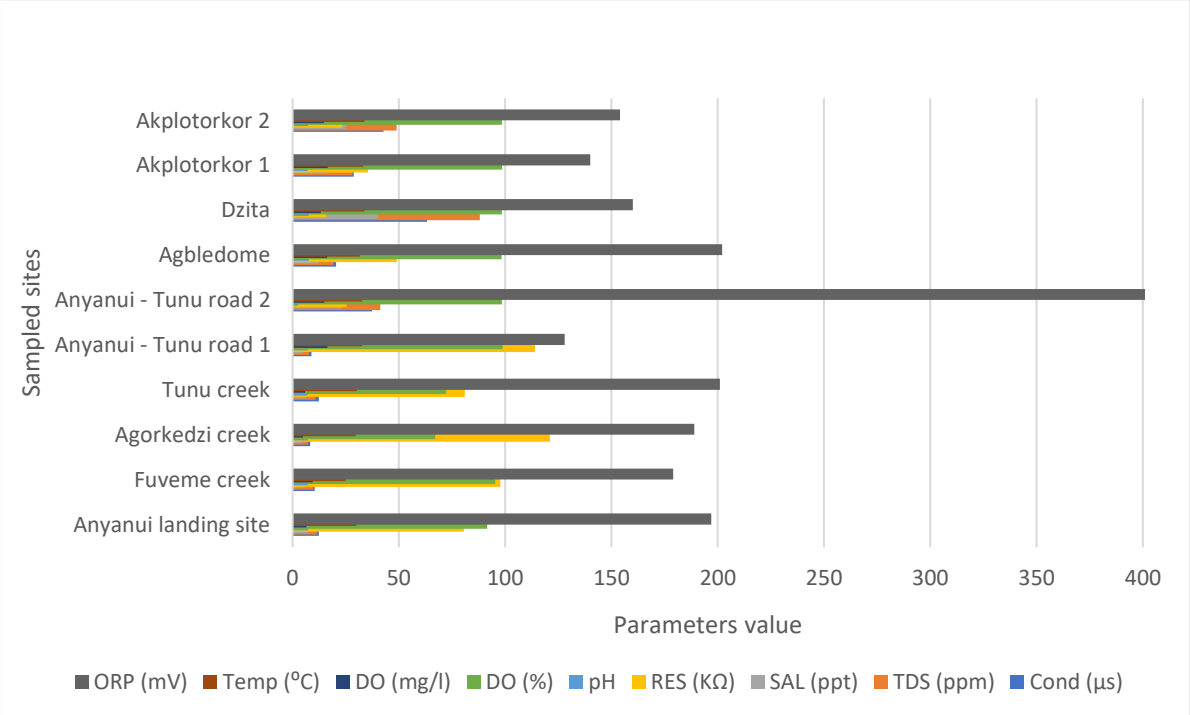

Figure 1: Temporal changes in physical and chemical water parameters at Anyanui-Dzita sampling stations.

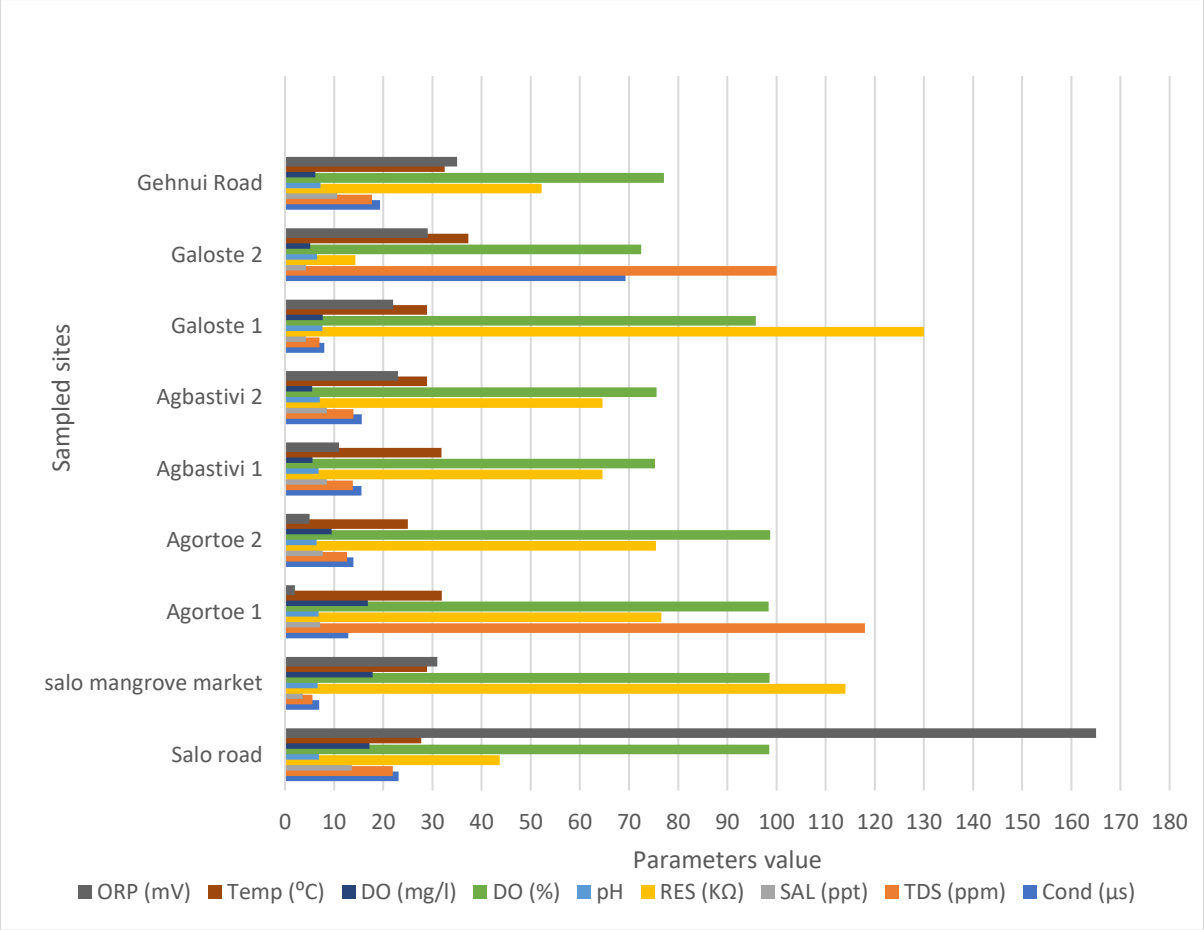

Figure 2: Temporal changes in physical and chemical water parameters at Saviatula – Galotse sampling stations.

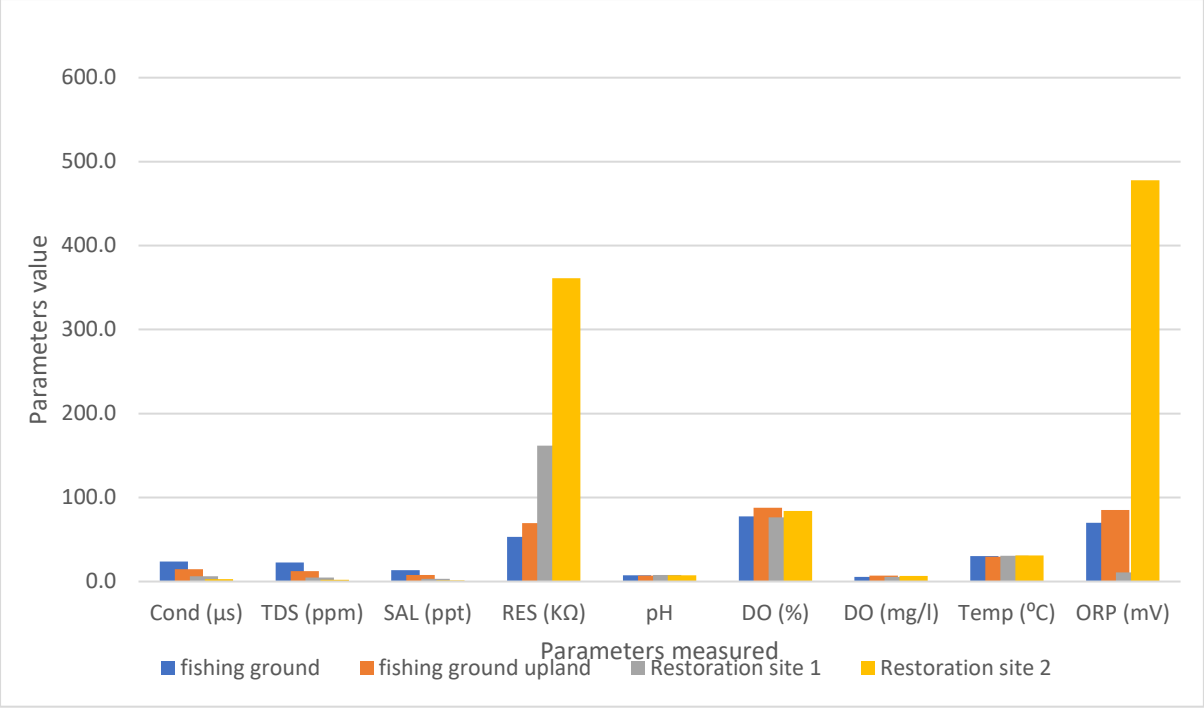

Figure 3: Temporal changes in physical and chemical water parameters at Aborlorvi-Norlopi sampling stations.

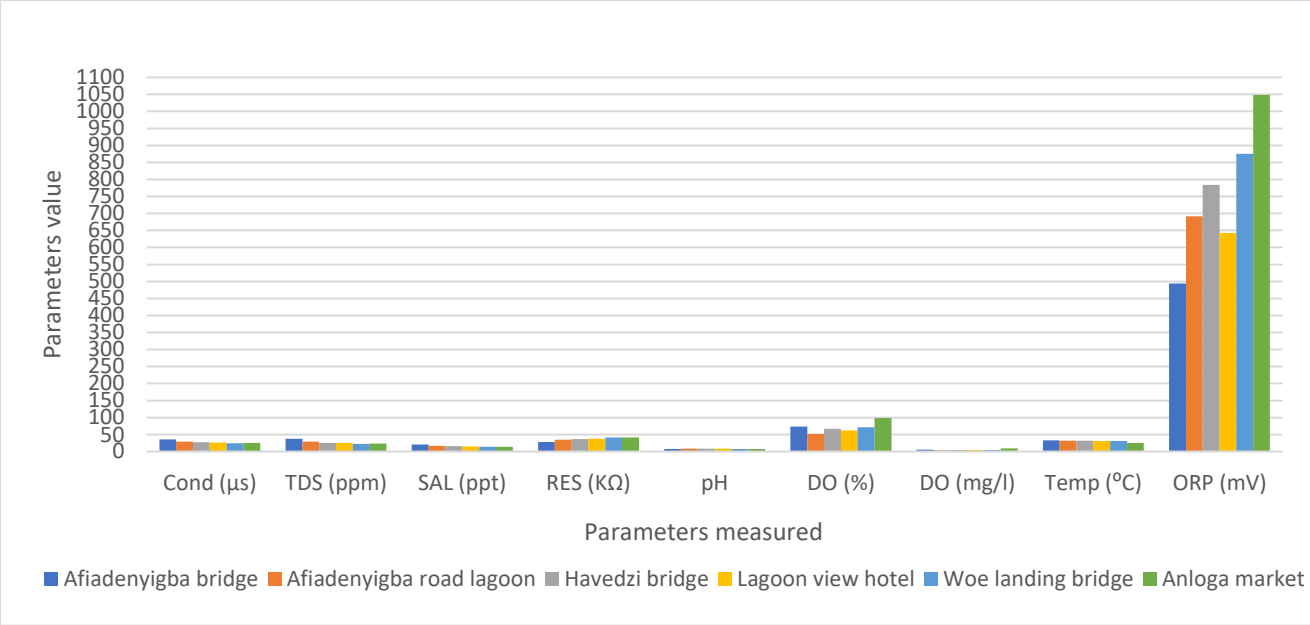

Figure 4: Temporal changes in physical and chemical water parameters at Afiadenyigba – Keta lagoon sampling stations.

3.3.2 Songor, Ada

Within the sampling site the highest value of conductivity occurred mostly within the creeks and the lagoon. The mean values recorded ranged between 31.4 and 50.4 µs. The highest value of 50.4 µs occurred at Big Ada bridge (Figure 7), whiles the lowest value of 31.4 µs was recorded at Patukope creek. However, values recorded outside the creek and the lagoon were relatively low. These were Azizanya, 6.23 µs and Ayigbo, 2.64 µs which were close to the values recorded within the main Volta River and the estuary.

Total dissolved solids (TDS), salinity and dissolved oxygen followed a similar trend as occurred in the lagoon and the creeks. The highest values of TDS occurred at Pute lagoon, 57.3 ppm, whiles the lowest value of 1.95 ppm was recorded at Ayigbo bridge (Figure 5). The value depicts the variation in dissolved solid that occurred in the main river, creeks, and the lagoon. Salinity values were higher in the lagoon and the creeks sampled, but relatively low in the main river. The highest salinity value in the lagoon, 28.6ppt occurred at Pute lagoon and 28.2ppt occurred in the Pute creek, the lagoon intake point. The lowest salinity value of 0.2ppt was recorded at Futue creek bridge that interconnect the main Volta River and the marshy area. The value probably was influenced by low sea tide and the rainfall during the sampling period.

Dissolved oxygen (DO) values were relatively uniform and high across the sampling stations with mean values ranging between 5.51 and 7.13 mg/l. This was probably due to the open lagoon with high wind, shallow depth and absence of significant algal mat on the sediment. The highest values of 6.09 mg/l and 6.91 mg/l were recorded at Futue and Luhue creek bridges, respectively.

The mean value of oxidation reduction potential (ORP) for the sampling site was between 1216 and 1547.3 mV. The highest values ranged between 832 – 1236 mV, and this occurred in the lagoon and the creeks sampled. The lowest values between 11 – 76.6 mV were recorded close to the estuary in the main Volta River.

Oxidation reduction potential (ORP) quantifies the water's capacity to act as either an oxidizing agent or a reducing agent. ORP is a measurement that indicates the ability of water to purify itself by breaking down waste materials and contaminants. A higher ORP value signifies a greater presence of oxygen in the water, which enhances the efficiency of bacteria that decompose dead tissue and pollutants. Therefore, higher positive values generally indicate better water quality and self-cleaning capacity. The quality of water in the wetlands is critical for biodiversity, livelihood sustenance and the health of the community members.

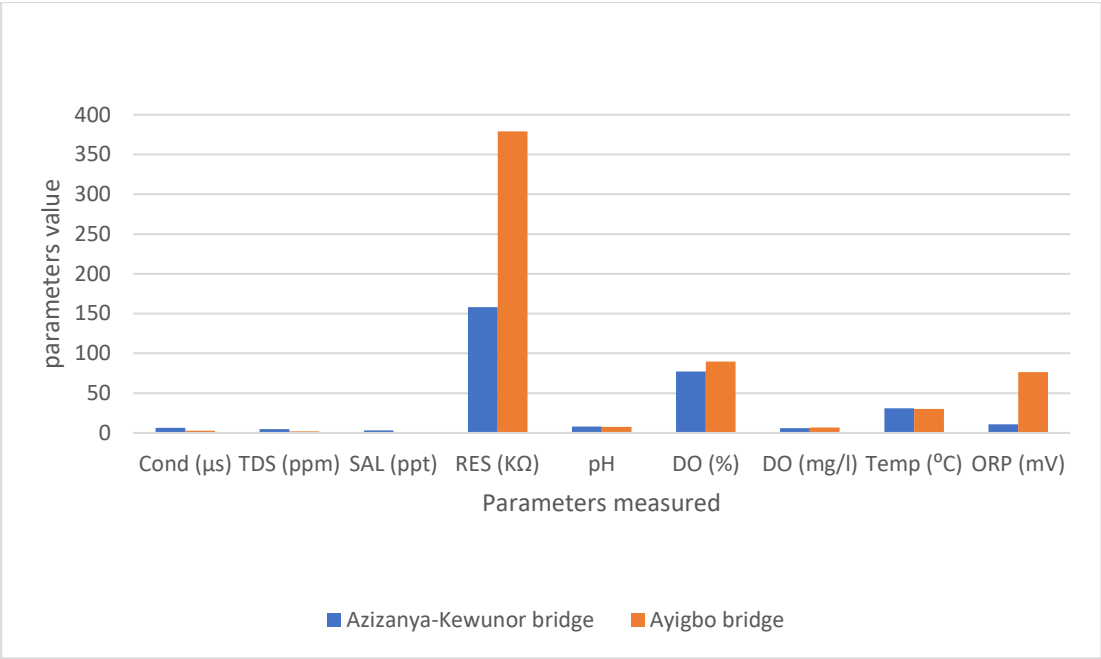

Figure 5: Temporal changes in physical and chemical water parameters at Azizanya-Ayigbo sampling stations.

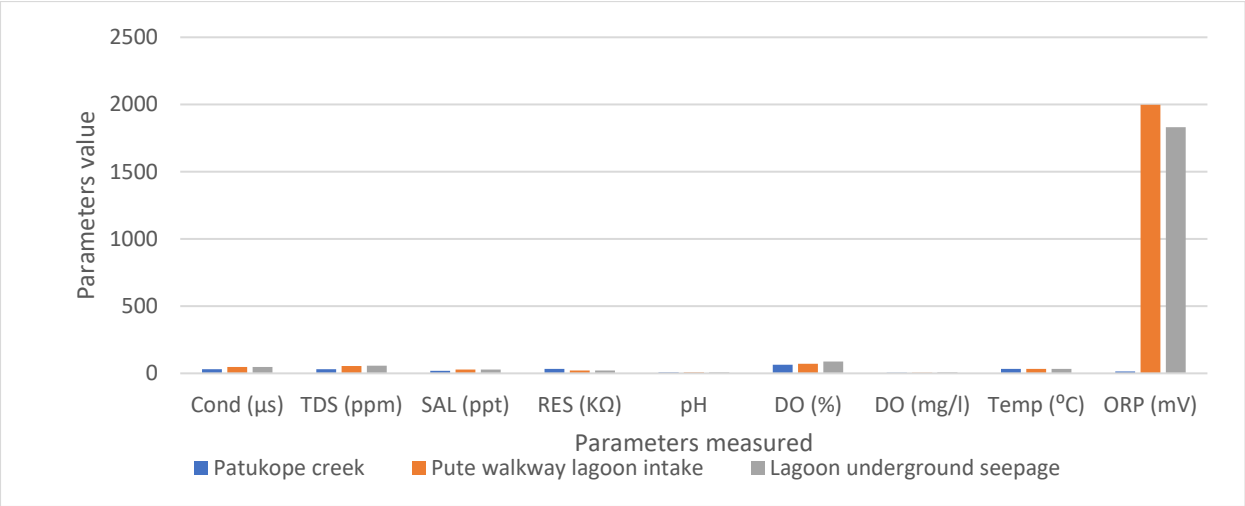

Figure 6: Temporal changes in physical and chemical water parameters at Patukope – Pute sampling stations.

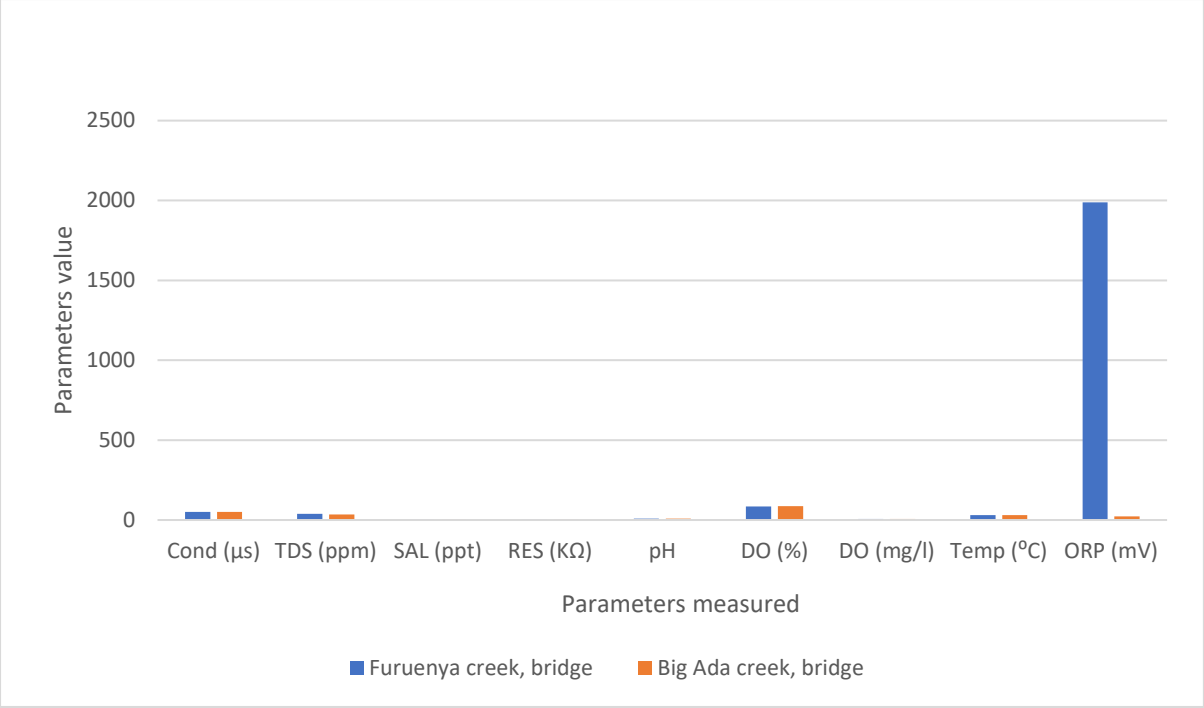

Figure 7: Temporal changes in physical and chemical water parameters at Futue – Luhue creeks.

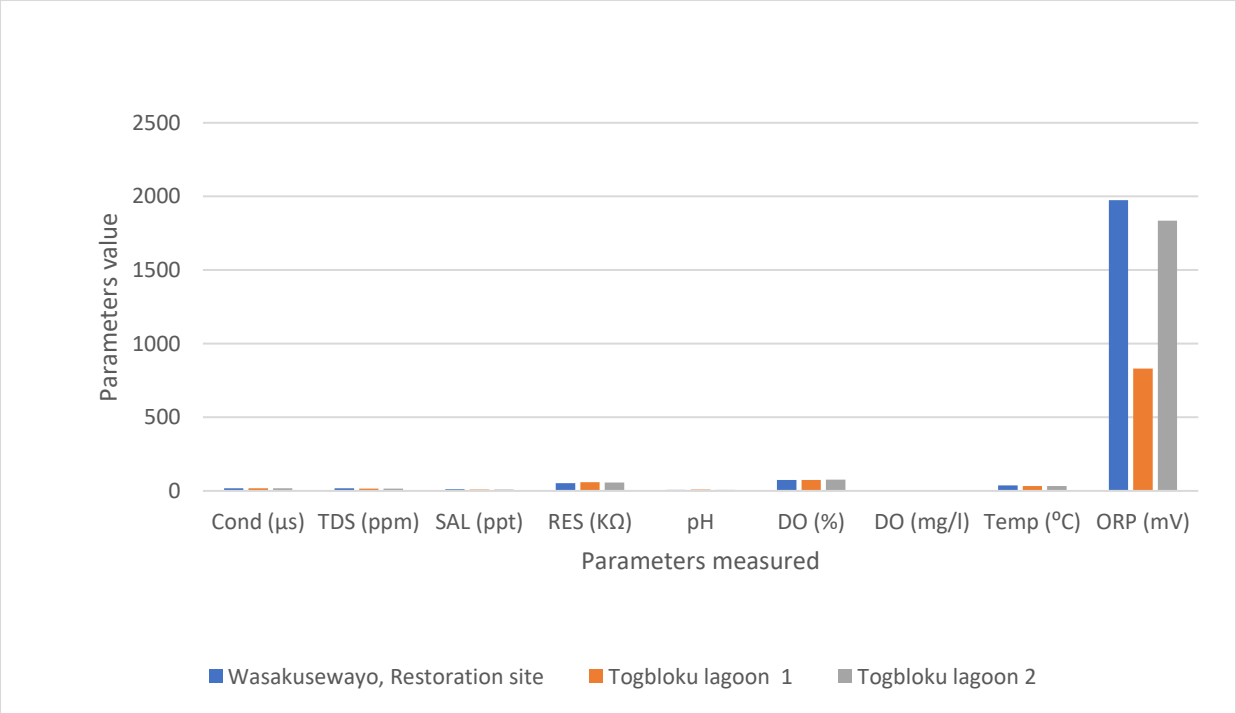

Figure 8: Temporal changes in physical and chemical water parameters at Wasakuse – Togbloku sampling stations.

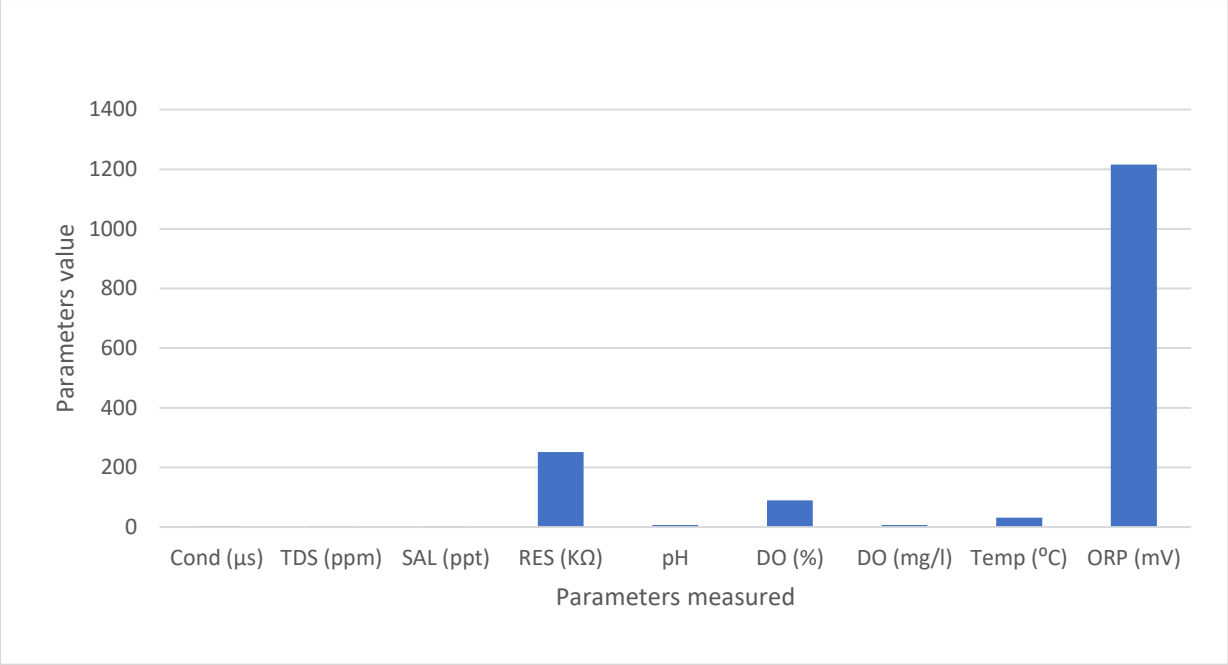

Figure 9: Temporal changes in physical and chemical water parameters at Obane creek.

3.4 Soil chemical and biological soil indicators

Soil health was assessed using various chemical and biological indicators. This was to ascertain its functionality to sustain diverse ecosystems for biodiversity, plant growth, nutrient retention and water regulation. The key indicators were nitrate, phosphate, potassium, soil pH, moisture, humus, and temperature.

3.4.1 Keta

The mean values of soil moisture and humus gathered were between 9.1 – 33% and 14.6 – 26.2% respectively. The level of other indicators varied within the sampling area. However, moisture and humus levels were comparatively uniform within the sampling site. Obviously, the moisture levels recorded were higher in the sampling stations closer to water bodies such as Aborlorvi, 25.4%, Afiadenyigba bridge, 26.4%, Genui, 38.7% and Galotse, 40.4%. The moisture levels were probably also influenced by the rainfall during the sampling period. Inversely, lower moisture values were recorded in sampling stations located upland or further away from water bodies.

It was observed that the mean humus value obtained, 14.6 – 26.2% was related to the presence of aquatic flora, litter fall and decomposed organic matter, especially within the mangrove stands. The values recorded were high at Tunu, 44.3%, Galotse, 40.6% and Agortoe, 37.5%, where dead mangroves, leaf falls and decomposed matter were observed. pH values recorded were at levels tolerable by most plants within the wetlands. The values recorded ranged between 5.55 and 7.0.

The level of potassium (K) recorded was high (56.25mg/kg) within the sampling area. However, the values recorded for nutrients varied within moist and dry sites. Potassium (K) is an essential element for crop growth, playing a key role in photosynthesis, protein synthesis, and many other essential functions in the plant. Soils frequently contain more than 20,000

ppm (parts per million) of potassium (K), but only a fraction of that is available for crop uptake. Soil factors that might impact K uptake include moisture, temperature and aeration. Nitrates, phosphates and potassium values recorded were moderately high in Agbledome, Agortoe, Agbastivi and Galotse. These were typically moist sites of the sampling area. Phosphate values recorded ranged between 1 and 32 mg/kg in the upland, whiles levels between 52 mg/kg and 103 mg/kg occurred in the moist areas. Adequate phosphate levels promote fruit, flower, and seed production, increase crop yields, promote root growth and hardiness of plants and hasten crop maturity.

Mean soil pH values obtained within the sampling site was between 6.2 and 7.0. Soil pH of 6 to 7.5 is ideal for the availability of phosphate for plant use. Values of less than 5.5 and 7.5 to 8.5 limit the availability of phosphate to plants. The levels of pH recorded within the site were within these ranges. Nitrate levels recorded did not vary within the upland sites, the values recorded ranged between 1mg/kg and 23mg/kg. The values obtained in the moist sites were between 37mg/kg and 73mg/kg.

The primary function of Nitrate (NO_3^-) is to serve as a source of nitrogen for the nutrition and growth of plants and soil microorganisms. Nitrate is usually deficient in acid soils; the optimum pH is between 6 and 8. Potassium (K) levels recorded were low to moderately high in uplands with values between 2 mg/kg and 65 mg/kg and the highest values between 78 and 201 mg/kg were recorded in the moist areas.

The mean potassium value recorded within the sampling site was between 42 and 117 mg/kg. Potassium (K) is an essential element that supports plant growth, playing a key role in photosynthesis, protein synthesis, and many other essential functions in the plant. Soils frequently contain more than 20,000 ppm (parts per million) of K, but only a fraction of that is available for crop uptake. Soil factors that might impact K uptake include moisture, temperature and aeration.

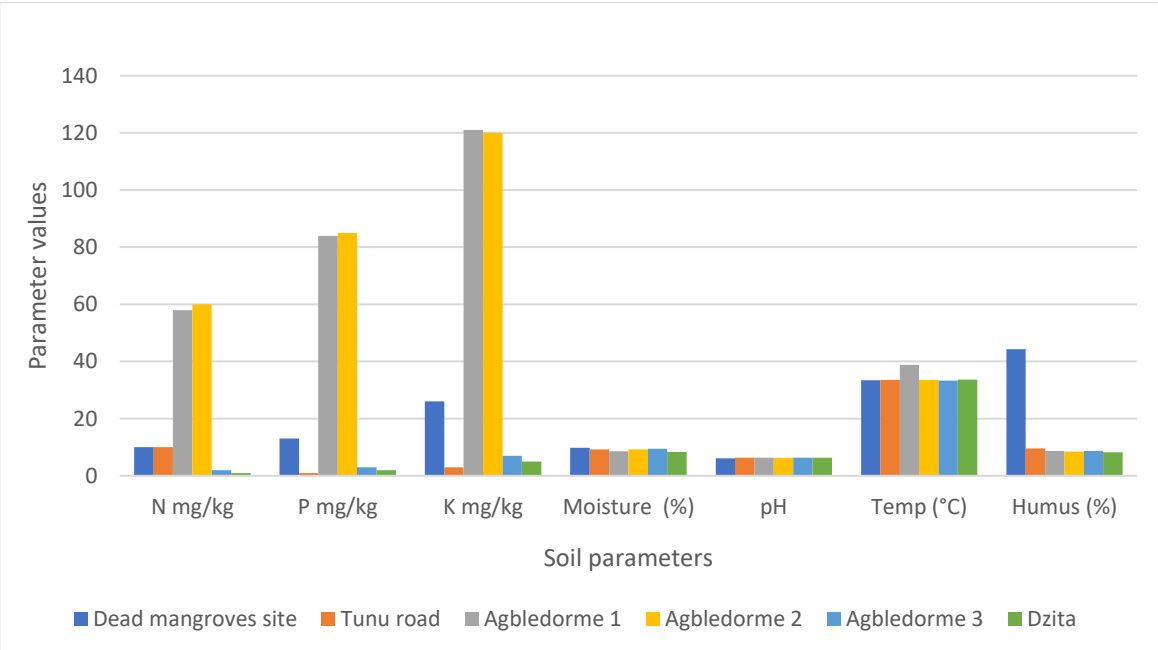

Figure 10: Chemical and biological soil parameters at Anyanui – Dzita sampling stations.

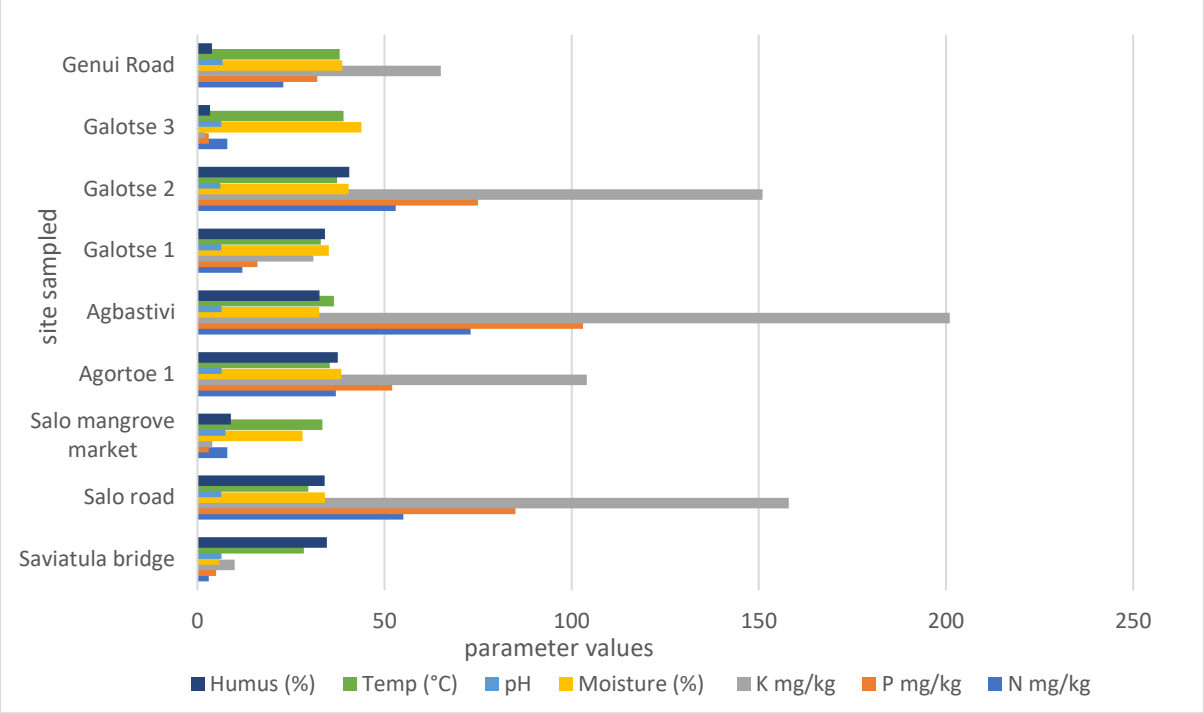

Figure 11: Chemical and biological soil parameters at Salo – Galotse sampling stations.

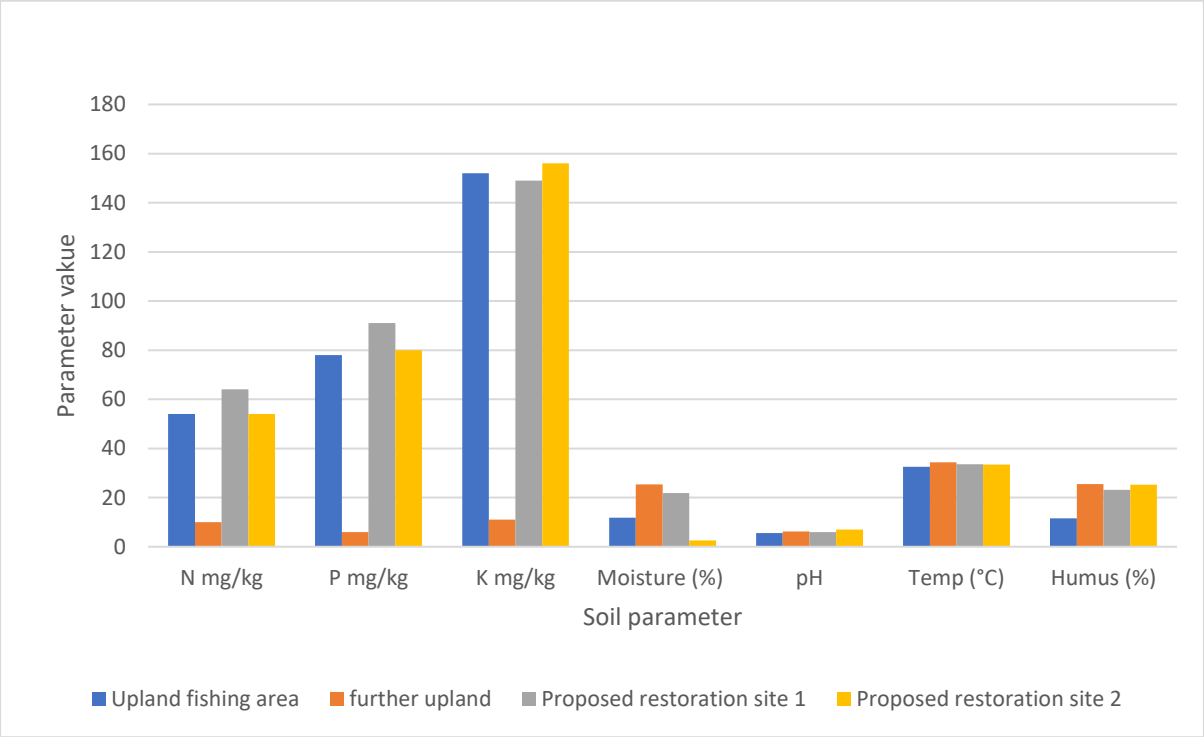

Figure 12: Chemical and biological soil parameters at Ablorlorvi sampling stations.

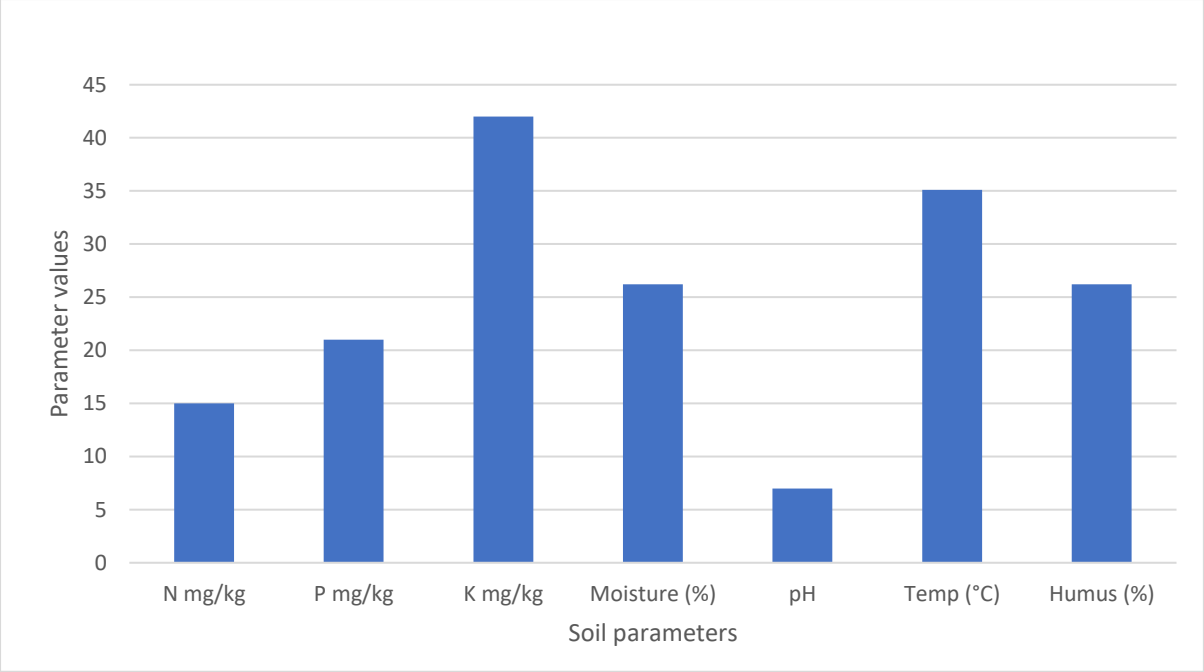

Figure 13: Chemical and biological soil parameters at Afiadenyigba sampling stations.

Figure 15: Testing water with change agents at Agblodorve

Figure 14: Testing water quality in the creek of Salo.

3.4.2 Songor, Ada

The mean moisture values recorded within the sampling site were between 21.5 and 29.45%. The levels recorded were high in Obane 25.1%, Wasakuse, 25.7%, Pute lagoon, 28.3%, and 30.6% at Pute lagoon intake. Probably an earlier rainfall and moist environmental conditions of the sampling site might have influenced the values obtained. Mean pH value obtained across the sampling site was between 5.57 and 6.84. An exceptional value of 4.78 was recorded at Pute lagoon.

Mean humus values recorded were between 16.9 and 26.0%. The value was however low (4.78%) at Pute lagoon intake. It was high at Obane restored mangrove site, 26%, but low at Pute lagoon, 9.2%. Humus is rich in nutrients and has a high surface area, which helps retain

moisture and improve soil structure. It is also essential for soil health as it provides a reservoir of nutrients, particularly nitrogen and phosphorus, which are critical for plant development.

Soil temperature recorded was high (38.4°C) at Pute walkway and low (34.2°C) at Obane restored mangrove site. The variation in the temperature values obtained may be due to weather conditions, soil moisture and humus content.

The levels of the soil nutrient values recorded were high within the sampling site, except at Pute lagoon. The mean nitrate (N) value obtained was between 48 mg/kg and 984 mg/kg. The value was high at Wasakusengwa (984 mg/kg) but low at Pute lagoon with a value of 3mg/kg. Phosphate value was high at Obane with value of 705 mg/kg but it was low (3 mg/kg) at Pute lagoon. Potassium value recorded was high (1,280 mg/kg) at Obane but low at Pute lagoon, 7 mg/kg. The high levels of nutrient values at Obane may probably be due to cattle grazing activities and the high levels of cow dung observed.

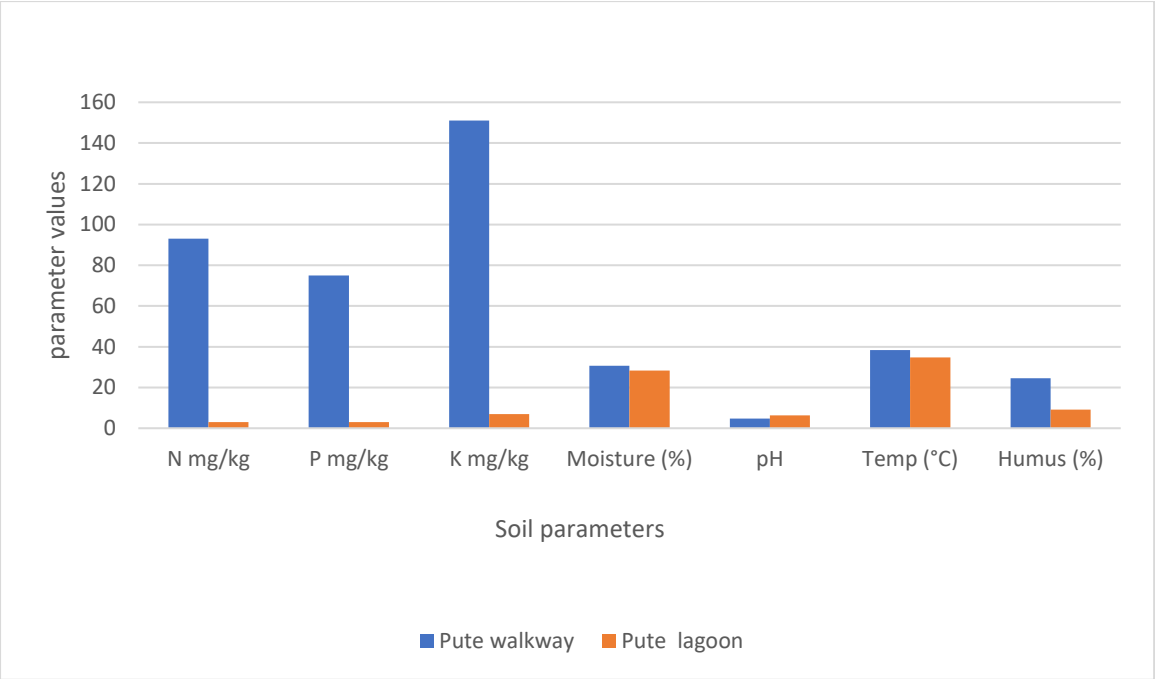

Figure 16: Chemical and biological soil parameters at Pute lagoon sampling stations.

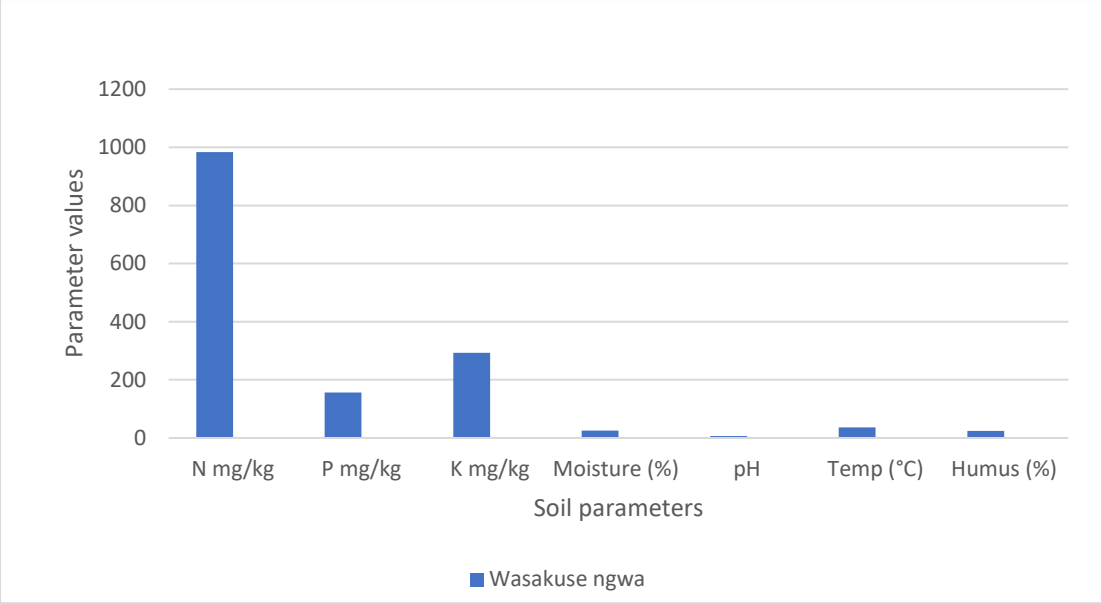

Figure 17: Chemical and biological soil parameters at Wasakuse-Troku sampling station.

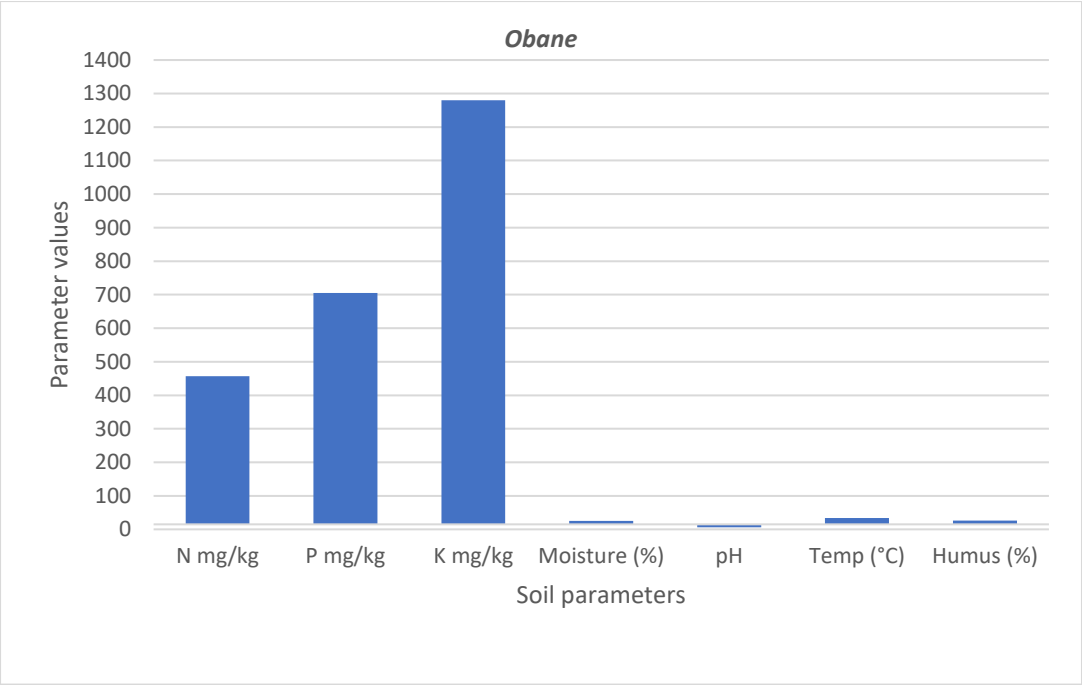

Figure 18: Chemical and biological soil parameters at Obane sampling station.

Figure 19: Testing soil at Patukope, Ada.

Figure 20: Moist mangrove habitats, Ada.

4.0 Conclusion

Wetland ecosystems are considered the most ecologically diverse; the nature of these ecosystems is also very dynamic. Therefore, the assessment of wetland ecosystem health is a very complicated task, because the impacts of the controlling factors vary spatially and temporally. Though seasonal changes were evident during the sampling period, the wetlands still remain productive ecosystems. They are vital ecosystems that provide numerous ecological, hydrological and socio-economic benefits and play crucial roles in biodiversity, water filtration and carbon sequestration. The water and soil that characterized the wetland ecosystem plays a fundamental role in shaping their structure and function. The level of the values of chemical and physical water and soil indicators sampled were optimal for development of flora and fauna activities. There were however, exceptional isolated cases in which levels of values recorded were critically low or high.

The water system has a way of buffering, but the fragile soil needs prudent and sustainable management measures to drastically reduce nutrient leaching. Aquatic plants mainly mangroves were well developed in sites with optimal water and soil characteristics. There was a clear linkage between the ecosystem and human wellbeing in terms of ecosystem services. Majority of the community members around the wetland depend on the wetland, especially the water resource as their livelihood. They are engaged either in agriculture or fishing activities. The over dependence on the wetlands' resources and ecosystem within the Ramsar sites constitute a major challenge for discussion and further studies.

5.0 Recommendations

It is recommended that the hydrological system within the site should be restored by opening the blocked water ways to enhance tidal exchange and improve aquatic life at the Anyanui estuary.

It is also imperative to effectively engage and collaborate with the communities to promote mangrove restoration and communal conservation activities.

Furthermore, long-term monitoring of water and soil indicator anomalies and nutrient levels is necessary to facilitate restoration of the disturbed habitat along the Anyanui creek and Tunu road.

6.0 References

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Appendix 1: Ecosystem Assessement – Physico-chemical water parameters.

Name of Site ……. Date …… Name of recorder ……

Coordinates	***Cond* (ms)**	***TDS (ppm)***	***SAL* (ppt)**	***RES (*K Ω)**	***pH***	***DO %***	***DO mg/l***	***ORP (mV)***	***Temp (*℃)**

Appendix 2: Ecosystem Assessment – Soil chemical and biological indicators

Name of Site ……. ***Date …….*** ***Name of recorder ……***

Coordinates	***N (mg/kg)***	***(P mg/kg)***	***K (mg/kg)***	***pH***	***Moisture (%)***	***Humus (%)***	***Temp (℃)***